SIMULATED HISTORICAL DATA

AGENTIC SOLUTIONS

GenBuddy

AI-Powered Backup Generator Maintenance Assistant

Turn generator telemetry, ATS events, and maintenance records into actionable readiness assessments — with a WorkBuddy AI agent that plans, investigates, and reasons on its own.

356	3	7	13
Dataset records	AI investigations	Agent reports	Planned steps (Inv 1)

02 / BUSINESS PROBLEM

A dark building is the first sign of trouble

ILLUSTRATIVE SCENARIO

Neighborhood Supermarket

A 250 m² store with two refrigerator cases, six freezer units, and a small backup generator in the back.

No on-site electrician. Visual inspections every quarter. ATS assumed to work.

What can go wrong between quarterly inspections?

- **Generator fails to start**
  Battery, fuel, or controller fault. Discovered only during the outage.
- **Generator starts but ATS does not transfer**
  Contactor or sensor fault. Generator runs but the building stays dark.
- **Generator overheats under sustained load**
  Marginal cooling capacity. Shuts down mid-outage.

THE GAP

SMEs have generators. They don't have engineers to interpret what the telemetry is telling them.

03 / OUR SOLUTION

GenBuddy turns raw signals into readiness and action

1 INPUTS

Existing generator data

- Telemetry (176 records, 13 params)
- ATS event log (176 records)
- Maintenance history (4 records)

→

2 AI INVESTIGATION

WorkBuddy LLM agent

- Plans a multi-step investigation
- Runs engineering tools
- Reads raw data independently
- Reasons about uncertainties
- Writes structured reports

→

3 OUTPUTS

Actionable insights

- Readiness verdict (READY / NOT READY)
- Prioritized work orders
- Root-cause timelines
- Contractor briefings

BUSINESS OUTCOME

A small business owner asks a plain-English question, and the agent returns a prioritized maintenance plan with evidence — ready for a qualified specialist to act on.

System architecture: deterministic engineering meets AI reasoning

E DETERMINISTIC ENGINEERING

Flask Dashboard
app.py · 12 API endpoints · templates/index.html

Engineering Scripts
validate_data.py (13 checks) · fault_investigation.py

CSV Data Sources
generator_data.csv · ats_events.csv · maintenance_history.csv

Request / Output
JSON files in requests/ · reports in outputs/

✓ Reproducible • testable • version-controlled

AI WORKBUDDY LLM AGENT

Plan
Builds a multi-step plan before executing

Execute Tools
Runs scripts, reads CSVs, queries API

Reason
Distinguishes observed facts from suspected causes

Write Reports
Outputs structured Markdown files for review

✓ Adaptive • investigative • auditable

FLOW	→	FILE	→	MANUAL	→	AGENT	→	OUTPUT
User question		JSON request		Paste to WorkBuddy		Plan + investigate		Reports in dashboard

One prompt, one goal, end-to-end execution

AGENT LOOP

1. **Plan** — Build multi-step plan before any tool runs
2. **Execute** — Run validation, scripts, reads, API
3. **Check** — Verify evidence against script findings
4. **Reason** — Distinguish observed vs suspected causes
5. **Report** — Write structured deliverables

PROOF: 13-STEP PLAN (INVESTIGATION 1)

1. Run validation script
2. Run fault investigation
3. Read script reports
4. Read maintenance history
5. Read generator specs
6. Read CSV data independently
7. Query Flask API for cross-verification
8. Examine 12 critical timestamps
9. Examine full thermal progression
10. Examine full ATS failure sequence
11. Cross-scenario comparison
12. Battery / fuel health
13. Synthesize findings

WORKBUDDY AGENT IN ACTION



Python scripts
validate_data.py · fault_investigation.py

Direct CSV reads + Flask API queries
independent verification + cross-check

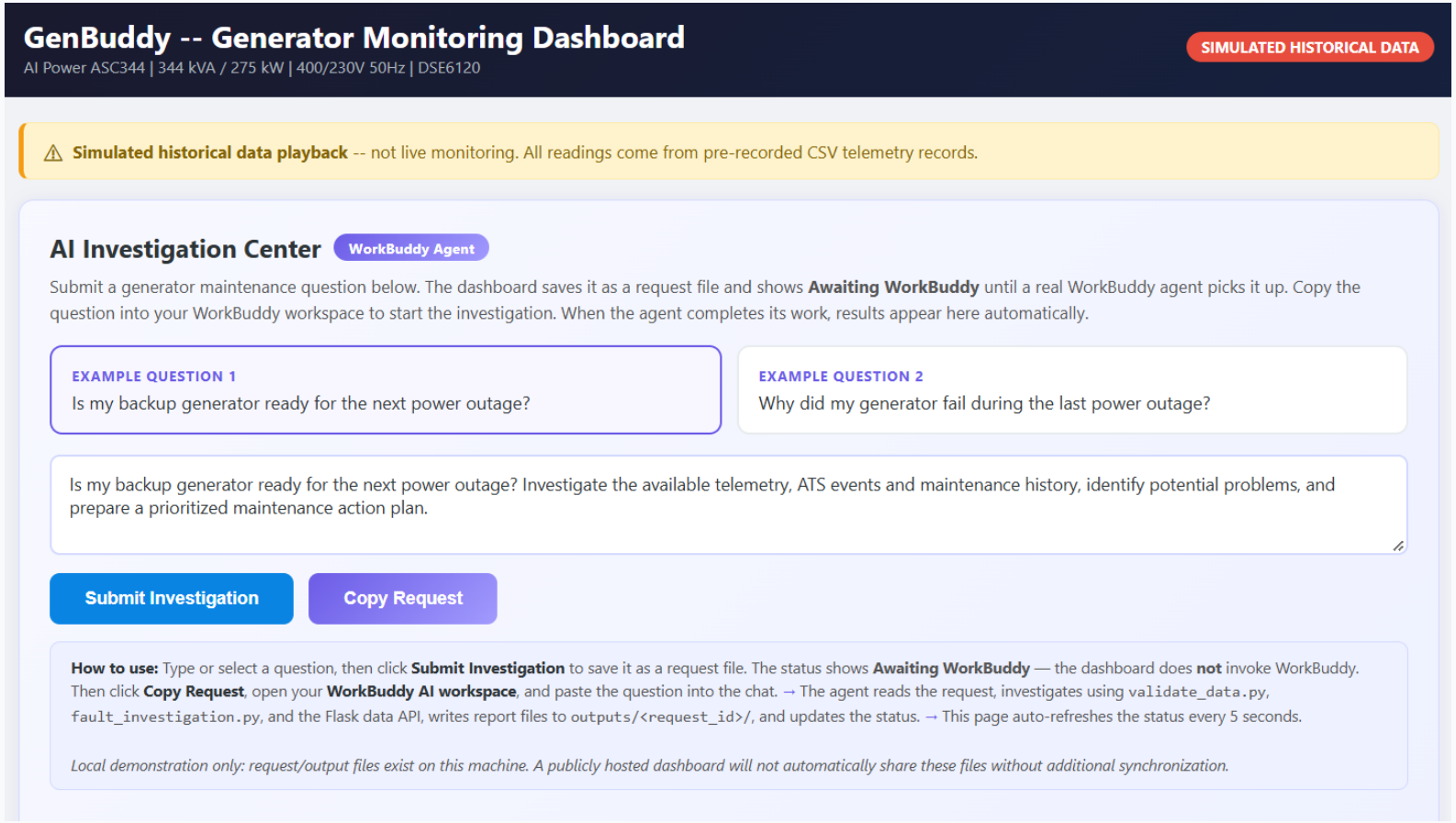
Report writes
7 AI reports · Markdown deliverables

13	20	3	7
planned steps (Inv 1)	tool calls (Inv 1)	investigations	AI-generated reports

06 / PRODUCT DEMO

The GenBuddy AI Investigation Center

A Flask dashboard where SMEs submit plain-English questions and read the agent's structured reports.



EXAMPLE QUESTIONS

What users actually ask

Is my backup generator ready for the next power outage?
INV 1 • READINESS

Why did my generator shut down during last week's outage?
INV 2 • ROOT CAUSE

What should I tell my maintenance contractor to fix?
INV 2 • CONTRACTOR BRIEF

Re-verify the readiness assessment end-to-end.
INV 3 • AUDIT

WHAT THE DASHBOARD DOES

Saves question to JSON · tracks status (awaiting → investigating → completed) · auto-polls every 5s · displays agent reports.

The agent found two real problems hiding in the data

SIMULATED DAY 2026-06-01 • THREE SCENARIOS • ALL DATA SIMULATED

✓ **Scenario A • 08:15**
Normal Operation

! **Scenario B • 11:15**
Overheating

× **Scenario C • 15:08**
ATS Transfer Anomaly

WO-01 • CRITICAL

ATS Transfer Anomaly

OBSERVED: 11 FAILED transfers in a row

OBSERVED: 0A current, position mismatch

SUSPECTED: contactor / actuator fault

⚠ Simulated data; outage duration inferred from alarm field, not independently measured

WO-02 • HIGH

Cooling System Margin

OBSERVED: coolant 44.8 → 107.7°C at 70% load

OBSERVED: oil 4.54 → 3.16 bar

SUSPECTED: blocked radiator / coolant aging

⚠ 105/107°C thresholds are simulation assumptions

WO-03 • INFO

Continue Scheduled Maint.

VERDICT: engine & alternator healthy

FUEL: 31.2 L/h matches spec

BATTERY: 24.27–28.04V healthy

+ add functional load tests to next service

CRITICAL
1 unresolved

HIGH
1 unresolved

VERDICT
NOT READY FOR SERVICE

The agent did NOT reuse the same plan for both investigations

1 INVESTIGATION 1
Readiness Assessment

"Is my backup generator ready for the next power outage?"

Question type	Forward-looking
Planned steps	13 steps
Data scope	All 356 records, 3 scenarios
Tools	Scripts + API + direct CSV reads
Analysis method	Cross-scenario comparison
Deliverable	Prioritized work orders
Maintenance angle	"Was it serviced recently?"

Key finding: 9.8 percentage points more rated loading → 22.8°C higher temperature

2 INVESTIGATION 2
Root Cause Analysis

"Why did my generator shut down during last week's outage?"

Question type	Backward-looking
Planned steps	8 steps (simplified)
Data scope	92 records, failure window only
Tools	Scripts + direct CSV reads (no API)
Analysis method	Timeline reconstruction
Deliverable	Contractor briefing
Maintenance angle	"Was the service adequate?"

Key finding: 0A current + 11 transfers = unsuccessful transfer

WHAT CHANGED — 7 DIMENSIONS OF ADAPTATION

Plan count | Data scope | Tool choice | Method | Format | Audience | Question framing

What we proved — and what we didn't

VERIFIED PROTOTYPE RESULTS

What GenBuddy could do

+ **Translate telemetry into plain-English flags**
  "11 failed ATS transfers" + "0A load current" → flagged for specialist investigation.
+ **Prioritize maintenance by business impact**
  CRITICAL · HIGH · INFORMATIONAL tags based on actual failure modes.
+ **Speak contractor fluently**
  Generates plain-language briefings with specific check instructions.
+ **Reason across scenarios**
  Cross-scenario comparison that fixed-temperature scripts cannot do.
+ **Self-correct during investigation**
  Agent fixed a temperature rise rate error in 3 report files.

HONEST LIMITATIONS

What we did not prove

! **No physical equipment connection**
  All data is simulated. DSE6120 thresholds are UNVERIFIED.
! **No validated real-world cost savings**
  No customer case studies; no measured downtime reduction.
! **Manual WorkBuddy triggering**
  User copies question from dashboard to WorkBuddy. No direct API.
! **Single-user, local runtime**
  File-based request system. No auth, no multi-user isolation.
! **Open investigation topics remain**
  DSE6120 thresholds · ambient temp · ATS manufacturer specs.

TRACK 2 PRINCIPLE **Show genuine WorkBuddy usage, not marketing claims.**

10 / CONCLUSION

GenBuddy = a backup generator that thinks with you

01

Plans before acting

The agent builds a multi-step plan, then executes tools and reads evidence independently.

02

Reasons, not just matches

Cross-scenario thermal comparison and timeline reconstruction go beyond pattern matching.

03

Adapts per question

Different plans, tools, scopes, and deliverables for readiness vs root cause investigations.

DEMO ACCESS • PLACEHOLDERS

GITHUB
https://github.com/cleo-blayea/Dormathon_Hackathon_Cleo.git

DEPLOYED APP
http://127.0.0.1:5000/

Thank you.